

CS306 MCQs - Part 1 (Topics 1-15, MCQs 1-50)

Question No: 1 (Marks: 1)

Which term best completes the following statement: 'o Square root of 100 o Value of pi o Result of `math.pow(2, 5)` Notes _____ CS306 introduction to python Noor institute of technology'?

- A. cs306
- B. 100
- C. notes

D. For

Explanation:

The correct answer, 'For', is directly supported by the provided text: 'o Square root of 100 o Value of pi o Result of `math.pow(2, 5)` Notes for CS306 introduction to python Noor institute of technology'.

Question No: 2 (Marks: 1)

Which term best completes the following statement: '_____ and use them in another script.'?

- A. use

B. Import

- C. script
- D. another

Explanation:

The correct answer, 'Import', is directly supported by the provided text: 'Import and use them in another script.'.

Question No: 3 (Marks: 1)

Which term best completes the following statement: 'Notes for CS306 introduction to _____'?

A. Python

- B. notes
- C. cs306
- D. introduction

Explanation:

The correct answer, 'Python', is directly supported by the provided text: 'Notes for CS306 introduction to python'.

Question No: 4 (Marks: 1)

Which term best completes the following statement: 'Summary ••••• Inheritance allows reuse of code from parent classes. Method Overriding lets you modify inherited behavior. The super() function accesses methods from _____ parent class. Polymorphism enables different classes to use the same method names with different implementations. Python supports multiple types of inheritance, including multiple and multilevel. ■ Homework / Practice'?

A. def

B. fly

C. The

D. overriding

Explanation:

The correct answer, 'The', is directly supported by the provided text: 'Summary ••••• Inheritance allows reuse of code from parent classes. Method Overriding lets you modify inherited behavior. The super() function accesses methods from the parent class. Polymorphism enables different classes to use the same method names with different implementations. Python supports multiple types of inheritance, including multiple and multilevel. ■ Homework / Practice'.

Question No: 5 (Marks: 1)

Which term best completes the following statement: 'o _____ child classes inherit from a single parent. ■ 6.'?

A. child

B. art

C. Multiple

D. def

Explanation:

The correct answer, 'Multiple', is directly supported by the provided text: 'o Multiple child classes inherit from a single parent. ■ 6.'.

Question No: 6 (Marks: 1)

Which term best completes the following statement: 'The isinstance() Function Used to check if an object belongs to a specific _____ or its subclass. print(isinstance(d, Dog)) print(isinstance(d, Animal)) # True # True ■ 9.'?

A. Class

B. airplane

C. multilevel

D. the

Explanation:

The correct answer, 'Class', is directly supported by the provided text: 'The isinstance() Function Used to check if an object belongs to a specific class or its subclass. print(isinstance(d, Dog)) print(isinstance(d, Animal)) # True # True ■ 9.'

Question No: 7 (Marks: 1)

Which term best completes the following statement: 'Polymorphism Polymorphism allows different classes to define methods with the same name, but possibly different behaviors. Example: class Bird: def fly(self): print("Bird can fly") class Airplane: def fly(self): print("Airplane flies in the sky") # Polymorphism using _____ def fly_test(entity): entity.fly() b = Bird() a = Airplane() fly_test(b) fly_test(a) # Output: Bird can fly # Output: Airplane flies in the sky ■ 8.'

- A. name
- B. object
- C. behavior

D. Function

Explanation:

The correct answer, 'Function', is directly supported by the provided text: 'Polymorphism Polymorphism allows different classes to define methods with the same name, but possibly different behaviors. Example: class Bird: def fly(self): print("Bird can fly") class Airplane: def fly(self): print("Airplane flies in the sky") # Polymorphism using function def fly_test(entity): entity.fly() b = Bird() a = Airplane() fly_test(b) fly_test(a) # Output: Bird can fly # Output: Airplane flies in the sky ■ 8.'

Question No: 8 (Marks: 1)

Which term best completes the following statement: 'Example: Multiple Inheritance class Father: def skills(self): print("Gardening, Programming") class Mother: def skills(self): print("Cooking, Art") class Child(Father, Mother): def skills(self): Father.skills(self) Mother.skills(self) print("Sports") c = Child() c.skills() Output: Gardening, Programming Cooking, Art Sports Notes for CS306 introduction to python _____ institute of technology ■ 7.'

- A. single

B. Noor

- C. possibly
- D. person

Explanation:

The correct answer, 'Noor', is directly supported by the provided text: 'Example: Multiple Inheritance class Father: def skills(self): print("Gardening, Programming") class Mother: def skills(self): print("Cooking, Art") class Child(Father, Mother): def skills(self): Father.skills(self) Mother.skills(self) print("Sports") c = Child() c.skills() Output: Gardening, Programming Cooking, Art Sports Notes for CS306 introduction to python Noor institute of technology ■ 7.'

Question No: 9 (Marks: 1)

Which term best completes the following statement: 'Real-World Example Problem: Model a Person class and derive a Student class that adds student-specific attributes and behavior. Code: class Person: def __init__(self, name): self.name = name def introduce(self): print(f'Hi, I'm {self.name}') class Student(Person): def __init__(self, name, student_id): super().__init__(name) self.student_id = student_id Notes for CS306 introduction to _____ Noor institute of technology def introduce(self): print(f'Hi, I'm {self.name}, my student ID is {self.student_id}') Usage: s = Student("Aisha", "VU1234") s.introduce() # Output: Hi, I'm Aisha, my student ID is VU1234 ■ 10.'

- A. object
- B. reuse
- C. classes

D. Python

Explanation:

The correct answer, 'Python', is directly supported by the provided text: 'Real-World Example Problem: Model a Person class and derive a Student class that adds student-specific attributes and behavior. Code: class Person: def __init__(self, name): self.name = name def introduce(self): print(f'Hi, I'm {self.name}') class Student(Person): def __init__(self, name, student_id): super().__init__(name) self.student_id = student_id Notes for CS306 introduction to python Noor institute of technology def introduce(self): print(f'Hi, I'm {self.name}, my student ID is {self.student_id}') Usage: s = Student("Aisha", "VU1234") s.introduce() # Output: Hi, I'm Aisha, my student ID is VU1234 ■ 10.'

Question No: 10 (Marks: 1)

Which term best completes the following statement: 'Binary files store data in a format that is not human-readable (e.g., images, videos). To open a binary file, use the 'b' mode in conjunction with other modes like 'rb' or 'wb'. _____ for CS306 introduction to python Noor institute of technology Example (Reading a Binary File): with open("image.jpg", "rb") as file: data = file.read() print(data) Example (Writing to a Binary File): with open("new_image.jpg", "wb") as file: file.write(data) ■ 9.'

- A. technology
- B. modifying

C. Notes

- D. concepts

Explanation:

The correct answer, 'Notes', is directly supported by the provided text: 'Binary files store data in a format that is not human-readable (e.g., images, videos). To open a binary file, use the 'b' mode in conjunction with other modes like 'rb' or 'wb'. Notes for CS306 introduction to python Noor institute of technology Example (Reading a Binary File): with open("image.jpg", "rb") as file: data = file.read() print(data) Example (Writing to a Binary File): with open("new_image.jpg", "wb") as file: file.write(data) ■ 9.'

Question No: 11 (Marks: 1)

Which term best completes the following statement: 'In _____ example, we will read from one file, modify its contents, and then write the updated content to a new file. Example: # Reading from the file with open("source.txt", "r") as file: lines = file.readlines() # Modifying the content lines.append("This is a new line.\n") # Writing to a new file with open("destination.txt", "w") as file: file.writelines(lines) ■■■ 8.'

- A. learned
- B. after

C. This

- D. automatically

Explanation:

The correct answer, 'This', is directly supported by the provided text: 'In this example, we will read from one file, modify its contents, and then write the updated content to a new file. Example: # Reading from the file with open("source.txt", "r") as file: lines = file.readlines() # Modifying the content lines.append("This is a new line.\n") # Writing to a new file with open("destination.txt", "w") as file: file.writelines(lines) ■■■ 8.'

Question No: 12 (Marks: 1)

Which term best completes the following statement: 'File Handling with Binary Files In addition to handling text files, Python also supports _____ file operations.'?

- A. manually
- B. exception

C. Binary

- D. read

Explanation:

The correct answer, 'Binary', is directly supported by the provided text: 'File Handling with Binary Files In addition to handling text files, Python also supports binary file operations.'

Question No: 13 (Marks: 1)

Which term best completes the following statement: 'Closing a File It is important to close a file after performing operations to ensure data is saved and resources are released. Syntax: file.close() Notes for CS306 introduction to _____ Noor institute of technology Failure to close a file can lead to memory leaks or corrupted data. ■■ 6.'?

- A. writing

B. Python

- C. image
- D. important

Explanation:

The correct answer, 'Python', is directly supported by the provided text: 'Closing a File It is important to close a file after performing operations to ensure data is saved and resources are released. Syntax: file.close() Notes for CS306 introduction to python Noor institute of technology Failure to close a file can lead to memory leaks or corrupted data. ■■ 6.'

Question No: 14 (Marks: 1)

Which term best completes the following statement: 'Summary ••••• File handling in Python involves opening, reading, writing, and closing files. You can open files in various modes, such as 'r', 'w', and 'a'. Use read(), readline(), and readlines() to read from a file. Use write() and writelines() to write to a file. Always close a file after operations or use the with statement _____ automatic file management. Binary file handling is done using 'rb' and 'wb' modes for reading and writing binary data. ■ Homework / Practice'?

A. content

B. For

C. handling

D. manually

Explanation:

The correct answer, 'For', is directly supported by the provided text: 'Summary ••••• File handling in Python involves opening, reading, writing, and closing files. You can open files in various modes, such as 'r', 'w', and 'a'. Use read(), readline(), and readlines() to read from a file. Use write() and writelines() to write to a file. Always close a file after operations or use the with statement for automatic file management. Binary file handling is done using 'rb' and 'wb' modes for reading and writing binary data. ■ Homework / Practice'.

Question No: 15 (Marks: 1)

Which term best completes the following statement: '_____ Handling Example Let's combine the concepts we've learned so far in a practical example.'?

A. File

B. binary

C. will

D. write

Explanation:

The correct answer, 'File', is directly supported by the provided text: 'File Handling Example Let's combine the concepts we've learned so far in a practical example.'.

Question No: 16 (Marks: 1)

Which term best completes the following statement: 'Using Context Manager (with Statement) _____ provides a convenient way to handle file operations using a context manager.'?

A. properly

B. supports

C. Python

D. handling

Explanation:

The correct answer, 'Python', is directly supported by the provided text: 'Using Context Manager (with Statement) Python provides a convenient way to handle file operations using a context manager.'.

Question No: 17 (Marks: 1)

Which term best completes the following statement: 'This ensures that the file is properly closed after its operations are completed, even if an _____ occurs. Syntax: with open("filename", "mode") as file: # File operations go here content = file.read() The context manager automatically handles the opening and closing of the file. Example: with open("example.txt", "r") as file: content = file.read() print(content) # No need to manually close the file ■ 7.'?

A. Exception

- B. important
- C. human
- D. closed

Explanation:

The correct answer, 'Exception', is directly supported by the provided text: 'This ensures that the file is properly closed after its operations are completed, even if an exception occurs. Syntax: with open("filename", "mode") as file: # File operations go here content = file.read() The context manager automatically handles the opening and closing of the file. Example: with open("example.txt", "r") as file: content = file.read() print(content) # No need to manually close the file ■ 7.'

Question No: 18 (Marks: 1)

Which term best completes the following statement: 'Working with Binary Files Binary files (e.g., images, audio files) require reading and writing in binary mode ("b"). Example: Reading a Binary File with open("image.jpg", "rb") as file: content = file.read() print(content) Example: Writing to a Binary File with open("output.jpg", "wb") as file: file.write(content) Notes for CS306 introduction to python _____ institute of technology ■ Summary • • • • Use open() to work with files in Python. Files can be opened in different modes ("r", "w", "a", etc.). Always close files after operations, or use the with statement to ensure automatic closure. Handle file exceptions using try and except blocks. ■ Homework / Practice 1. 2. 3.'?

A. Noor

- B. image
- C. readlines
- D. relative

Explanation:

The correct answer, 'Noor', is directly supported by the provided text: 'Working with Binary Files Binary files (e.g., images, audio files) require reading and writing in binary mode ("b"). Example: Reading a Binary File with open("image.jpg", "rb") as file: content = file.read() print(content) Example: Writing to a Binary File with open("output.jpg", "wb") as file: file.write(content) Notes for CS306 introduction to python Noor institute of technology ■ Summary • • • • Use open() to work with files in Python. Files can be opened in different modes ("r", "w", "a", etc.). Always close files after operations, or use the with statement to ensure automatic closure. Handle file exceptions using try and except blocks. ■ Homework / Practice 1. 2. 3.'

Question No: 19 (Marks: 1)

Which term best completes the following statement: 'Working with File _____ When opening a file, you can specify a relative path (from the current directory) or an absolute path (full path from the root). Example: file = open("/home/user/example.txt", "r") # Absolute path ■ 8.'?

A. Paths

- B. even
- C. before
- D. images

Explanation:

The correct answer, 'Paths', is directly supported by the provided text: 'Working with File Paths When opening a file, you can specify a relative path (from the current directory) or an absolute path (full path from the root). Example: file = open("/home/user/example.txt", "r") # Absolute path ■ 8.'

Question No: 20 (Marks: 1)

Which term best completes the following statement: '_____ Files It's crucial to always close a file after performing operations to free up system resources. Syntax: file.close() ■■ 6.'?

A. Closing

- B. institute
- C. are
- D. images

Explanation:

The correct answer, 'Closing', is directly supported by the provided text: 'Closing Files It's crucial to always close a file after performing operations to free up system resources. Syntax: file.close() ■■ 6.'

Question No: 21 (Marks: 1)

Which term best completes the following statement: 'Opening Files in Python Before performing any operations on a file (like reading or writing), you need to open the file using the open() function. Notes for CS306 introduction to _____ Noor institute of technology Syntax: file_object = open("filename", "mode") File Modes: "r": Read (default mode).'?

A. when

B. Python

- C. content
- D. homework

Explanation:

The correct answer, 'Python', is directly supported by the provided text: 'Opening Files in Python Before performing any operations on a file (like reading or writing), you need to open the file using the open() function. Notes for CS306 introduction to python Noor institute of technology Syntax: file_object = open("filename", "mode") File Modes: "r": Read (default mode).'

Question No: 22 (Marks: 1)

Which term best completes the following statement: 'Writing to Files To write to a file, open it in write ("w") or append ("a") mode. Notes for CS306 introduction to python Noor institute of technology write(): Writes the string to the file (overwrites the file if it exists). file = open("example.txt", "w") file.write("Hello, World!\nThis is a test.") file.close() writelines(): Writes a _____ of strings to a file. file = open("example.txt", "w") lines = ["First line\n", "Second line\n"] file.writelines(lines) file.close() ■ 5.?'

- A. already
- B. audio
- C. before

D. List

Explanation:

The correct answer, 'List', is directly supported by the provided text: 'Writing to Files To write to a file, open it in write ("w") or append ("a") mode. Notes for CS306 introduction to python Noor institute of technology write(): Writes the string to the file (overwrites the file if it exists). file = open("example.txt", "w") file.write("Hello, World!\nThis is a test.") file.close() writelines(): Writes a list of strings to a file. file = open("example.txt", "w") lines = ["First line\n", "Second line\n"] file.writelines(lines) file.close() ■ 5.'

Question No: 23 (Marks: 1)

Which term best completes the following statement: 'Context Manager (with Statement) Using with statement helps ensure that the file is properly closed after the operations, even if an error occurs. Example: with open("_____.txt", "r") as file: content = file.read() print(content) # No need to call file.close(), it's automatically handled ■ 10.?'

- A. existing

B. Example

- C. permission
- D. exist

Explanation:

The correct answer, 'Example', is directly supported by the provided text: 'Context Manager (with Statement) Using with statement helps ensure that the file is properly closed after the operations, even if an error occurs. Example: with open("example.txt", "r") as file: content = file.read() print(content) # No need to call file.close(), it's automatically handled ■ 10.'

Question No: 24 (Marks: 1)

Which term best completes the following statement: 'Introduction to File Handling File handling in Python allows you to interact with _____s—read from them, write to them, and manipulate file contents.'?

- A. handling
- B. print

C. File

D. handle

Explanation:

The correct answer, 'File', is directly supported by the provided text: 'Introduction to File Handling File handling in Python allows you to interact with files—read from them, write to them, and manipulate file contents.'

Question No: 25 (Marks: 1)

Which term best completes the following statement: 'Opens the file for appending data. "b": Binary mode (used for non-text files like images). "x": _____ creation.'?

A. open

B. except

C. Exclusive

D. methods

Explanation:

The correct answer, 'Exclusive', is directly supported by the provided text: 'Opens the file for appending data. "b": Binary mode (used for non-text files like images). "x": Exclusive creation.'

Question No: 26 (Marks: 1)

Which term best completes the following statement: '_____ to Python Libraries In Python, a library is a collection of pre-written code that provides functionality for various tasks like file I/O, math operations, system operations, and web development.'?

A. file

B. pip

C. provides

D. Introduction

Explanation:

The correct answer, 'Introduction', is directly supported by the provided text: 'Introduction to Python Libraries In Python, a library is a collection of pre-written code that provides functionality for various tasks like file I/O, math operations, system operations, and web development.'

Question No: 27 (Marks: 1)

Which term best completes the following statement: 'Libraries help avoid writing repetitive code and make programming faster and more efficient. Two types of _____:'?

A. programming

B. tasks

C. about

D. Libraries

Explanation:

The correct answer, 'Libraries', is directly supported by the provided text: 'Libraries help avoid writing repetitive code and make programming faster and more efficient. Two types of libraries:'.

Question No: 28 (Marks: 1)

Which term best completes the following statement: '■ _____ Objectives •••• Understand what libraries are in Python. Learn how to import and use external libraries in Python. Explore some commonly used standard libraries in Python. Learn about third-party libraries and how to install them using pip. ■ 1.'?

- A. how
- B. written
- C. that

D. Lecture

Explanation:

The correct answer, 'Lecture', is directly supported by the provided text: '■ Lecture Objectives •••• Understand what libraries are in Python. Learn how to import and use external libraries in Python. Explore some commonly used standard libraries in Python. Learn about third-party libraries and how to install them using pip. ■ 1.'.

Question No: 29 (Marks: 1)

Which term best completes the following statement: 'When you create an object, Python calls the class constructor (___init___ method) to initialize the object's attributes. _____ an Object: person1 = Person("Alice", 30) person2 = Person("Bob", 25) Now, you can access the attributes and methods of the objects: print(person1.name) # Output: Alice person2.greet() # Output: Hello, my name is Bob and I am 25 years old. ■■ 3.'?

A. Creating

- B. concept
- C. hello
- D. around

Explanation:

The correct answer, 'Creating', is directly supported by the provided text: 'When you create an object, Python calls the class constructor (___init___ method) to initialize the object's attributes. Creating an Object: person1 = Person("Alice", 30) person2 = Person("Bob", 25) Now, you can access the attributes and methods of the objects: print(person1.name) # Output: Alice person2.greet() # Output: Hello, my name is Bob and I am 25 years old. ■■ 3.'.

Question No: 30 (Marks: 1)

Which term best completes the following statement: 'OOP makes it easier to create modular, reusable, and maintainable code. Key Concepts of OOP: Classes: Blueprints for creating objects. Objects: _____ of a class. Attributes: Variables that belong to a class or object. Methods: Functions that belong to a class or object. •••• ■■ 2.'?

A. concept

B. Instances

C. can

D. variables

Explanation:

The correct answer, 'Instances', is directly supported by the provided text: 'OOP makes it easier to create modular, reusable, and maintainable code. Key Concepts of OOP: Classes: Blueprints for creating objects. Objects: Instances of a class. Attributes: Variables that belong to a class or object. Methods: Functions that belong to a class or object. ••• ■■ 2.'

Question No: 31 (Marks: 1)

Which term best completes the following statement: 'It defines the attributes and methods common to all objects of that class. Syntax to Define a Class: class ClassName: # constructor (__init__ method) def __init__(self, attribute1, attribute2): self.attribute1 = attribute1 self.attribute2 = attribute2 # method def method_name(self): pass Example: class Person: def __init__(self, name, age): self.name = name self.age = age def greet(self): print(f"_____, my name is {self.name} and I am {self.age} years old.") Notes for CS306 introduction to python Noor institute of technology Objects An object is an instance of a class.'?

A. cs306

B. Hello

C. technology

D. inheritance

Explanation:

The correct answer, 'Hello', is directly supported by the provided text: 'It defines the attributes and methods common to all objects of that class. Syntax to Define a Class: class ClassName: # constructor (__init__ method) def __init__(self, attribute1, attribute2): self.attribute1 = attribute1 self.attribute2 = attribute2 # method def method_name(self): pass Example: class Person: def __init__(self, name, age): self.name = name self.age = age def greet(self): print(f"Hello, my name is {self.name} and I am {self.age} years old.") Notes for CS306 introduction to python Noor institute of technology Objects An object is an instance of a class.'

Question No: 32 (Marks: 1)

Which term best completes the following statement: 'Defining Classes and Objects Classes A class is a blueprint for creating _____.'

A. oop

B. four

C. hello

D. Objects

Explanation:

The correct answer, 'Objects', is directly supported by the provided text: 'Defining Classes and Objects Classes A class is a blueprint for creating objects.'

Question No: 33 (Marks: 1)

Which term best completes the following statement: '■ Lecture Objectives • Understand the concept of Object-Oriented Programming (OOP). Notes for CS306 introduction to python _____ institute of technology Learn about the four main principles of OOP: Encapsulation, Abstraction, Inheritance, and Polymorphism. Understand how to implement these principles in Python using classes and objects. Learn about methods and attributes in Python. • • • ■■■ 1.'?

A. attribute2

B. Noor

C. four

D. creating

Explanation:

The correct answer, 'Noor', is directly supported by the provided text: '■ Lecture Objectives • Understand the concept of Object-Oriented Programming (OOP). Notes for CS306 introduction to python Noor institute of technology Learn about the four main principles of OOP: Encapsulation, Abstraction, Inheritance, and Polymorphism. Understand how to implement these principles in Python using classes and objects. Learn about methods and attributes in Python. • • • ■■■ 1.'.

Question No: 34 (Marks: 1)

Which term best completes the following statement: '_____ to Object-Oriented Programming (OOP) Object-Oriented Programming (OOP) is a programming paradigm that organizes software design around objects and classes.'?

A. constructor

B. Introduction

C. functions

D. defining

Explanation:

The correct answer, 'Introduction', is directly supported by the provided text: 'Introduction to Object-Oriented Programming (OOP) Object-Oriented Programming (OOP) is a programming paradigm that organizes software design around objects and classes.'.

Question No: 35 (Marks: 1)

Which term best completes the following statement: 'Introduction to File Handling File _____ in Python allows you to read from and write to files.'?

A. operations

B. example

C. opened

D. Handling

Explanation:

The correct answer, 'Handling', is directly supported by the provided text: 'Introduction to File Handling File handling in Python allows you to read from and write to files.'

Question No: 36 (Marks: 1)

Which term best completes the following statement: 'Data is added at the end. 'b': Binary mode (used with other modes like 'rb' or 'wb' for binary files). 'x': _____ creation.'?

- A. files
- B. built
- C. safe

D. Exclusive

Explanation:

The correct answer, 'Exclusive', is directly supported by the provided text: 'Data is added at the end. 'b': Binary mode (used with other modes like 'rb' or 'wb' for binary files). 'x': Exclusive creation.'

Question No: 37 (Marks: 1)

Which term best completes the following statement: 'Creates a new file or truncates an existing file. 'a': _____.'?

- A. common
- B. operation
- C. introduction

D. Append

Explanation:

The correct answer, 'Append', is directly supported by the provided text: 'Creates a new file or truncates an existing file. 'a': Append.'

Question No: 38 (Marks: 1)

Which term best completes the following statement: 'This function returns a file object, which provides methods and attributes to perform file operations. _____: file_object = open("filename", "mode") Parameters: • • "filename": The name of the file to be opened (can include the file path). "mode": The mode in which the file is opened (read, write, append, etc.). Modes of File Operations: • • • • • 'r': Read (default mode).'?

- A. exclusive
- B. efficiently
- C. with

D. Syntax

Explanation:

The correct answer, 'Syntax', is directly supported by the provided text: 'This function returns a file object, which provides methods and attributes to perform file operations. Syntax: file_object = open("filename", "mode") Parameters: • • "filename": The name of the file to be opened (can include the

file path). "mode": The mode in which the file is opened (read, write, append, etc.). Modes of File Operations: ••••• 'r': Read (default mode).'

Question No: 39 (Marks: 1)

Which term best completes the following statement: 'Opens the file for reading. 'w': _____'?

- A. for
- B. methods
- C. opened

D. Write

Explanation:

The correct answer, 'Write', is directly supported by the provided text: 'Opens the file for reading. 'w': Write.'

Question No: 40 (Marks: 1)

Which term best completes the following statement: 'Opening a File _____ performing any operation on a file, you must open it using Python's built-in open() function.'?

- A. log

B. Before

- C. file
- D. functions

Explanation:

The correct answer, 'Before', is directly supported by the provided text: 'Opening a File Before performing any operation on a file, you must open it using Python's built-in open() function.'

Question No: 41 (Marks: 1)

Which term best completes the following statement: '■ _____ Objectives • Understand the concept of file handling in Python. Notes for CS306 introduction to python Noor institute of technology ••• Learn how to open, read, write, and close files in Python. Explore various modes for file operations in Python. Learn about context managers for safe file handling. ■ 1.'?

A. Lecture

- B. contents
- C. use
- D. new

Explanation:

The correct answer, 'Lecture', is directly supported by the provided text: '■ Lecture Objectives • Understand the concept of file handling in Python. Notes for CS306 introduction to python Noor institute of technology ••• Learn how to open, read, write, and close files in Python. Explore various modes for file operations in Python. Learn about context managers for safe file handling. ■ 1.'

Question No: 42 (Marks: 1)

Which term best completes the following statement: 'Python provides several built-in functions for handling files efficiently. Common File Operations: •••• Opening a file. Reading from a file. Writing to a file. _____ a file. ■■ 2.'?

A. modes

B. Closing

C. open

D. once

Explanation:

The correct answer, 'Closing', is directly supported by the provided text: 'Python provides several built-in functions for handling files efficiently. Common File Operations: •••• Opening a file. Reading from a file. Writing to a file. Closing a file. ■■ 2.'.

Question No: 43 (Marks: 1)

Which term best completes the following statement: '_____ to Exceptions An exception is an event that disrupts the normal flow of a program.'?

A. event

B. when

C. are

D. Introduction

Explanation:

The correct answer, 'Introduction', is directly supported by the provided text: 'Introduction to Exceptions An exception is an event that disrupts the normal flow of a program.'.

Question No: 44 (Marks: 1)

Which term best completes the following statement: '_____ Python encounters an error, it raises an exception to indicate that something went wrong. Types of Errors in Python:'?

A. When

B. wrong

C. exception

D. using

Explanation:

The correct answer, 'When', is directly supported by the provided text: 'When Python encounters an error, it raises an exception to indicate that something went wrong. Types of Errors in Python:'.

Question No: 45 (Marks: 1)

Which term best completes the following statement: '■ Lecture Objectives •••• Understand what exceptions are and why they occur. _____ how to handle exceptions in Python using try, except, else, and finally. Explore common built-in exceptions and how to raise custom exceptions. Understand the importance of exception handling in making Python programs more robust. Notes for CS306 introduction to python Noor institute of technology ■■ 1.'?

A. Learn

- B. except
- C. built
- D. understand

Explanation:

The correct answer, 'Learn', is directly supported by the provided text: '■ Lecture Objectives •••• Understand what exceptions are and why they occur. Learn how to handle exceptions in Python using try, except, else, and finally. Explore common built-in exceptions and how to raise custom exceptions. Understand the importance of exception handling in making Python programs more robust. Notes for CS306 introduction to python Noor institute of technology ■■ 1.'.

Question No: 46 (Marks: 1)

Which term best completes the following statement: 'Classes and Objects in Python Creating a Class: class _____: pass This defines an empty class named Student. Creating an Object: s1 = Student() Here, s1 is an object (or instance) of the Student class. ■■ 3.'?

A. Student

- B. perform
- C. variable
- D. return

Explanation:

The correct answer, 'Student', is directly supported by the provided text: 'Classes and Objects in Python Creating a Class: class Student: pass This defines an empty class named Student. Creating an Object: s1 = Student() Here, s1 is an object (or instance) of the Student class. ■■ 3.'.

Question No: 47 (Marks: 1)

Which term best completes the following statement: 'Adding Methods to a Class _____ can add multiple methods to perform actions related to the object. Example: class Calculator: def __init__(self, a, b): self.a = a self.b = b def add(self): return self.a + self.b def subtract(self): return self.a - self.b Using Methods: calc = Calculator(10, 5) print(calc.add()) # Output: 15 print(calc.subtract()) # Output: 5 ■ 6.'?

- A. what
- B. however

C. You

- D. outside

Explanation:

The correct answer, 'You', is directly supported by the provided text: 'Adding Methods to a Class You can add multiple methods to perform actions related to the object. Example: class Calculator: def __init__(self, a, b): self.a = a self.b = b def add(self): return self.a + self.b def subtract(self): return self.a - self.b Using Methods: calc = Calculator(10, 5) print(calc.add()) # Output: 15 print(calc.subtract()) # Output: 5 ■ 6.'

Question No: 48 (Marks: 1)

Which term best completes the following statement: 'It is used to initialize attributes. Example: class Student: def __init__(self, name, roll_no): self.name = name self.roll_no = roll_no Creating Objects: s1 = Student("Ali", 123) s2 = Student("Sara", 456) Accessing Attributes: _____(s1.name) print(s2.roll_no) # Output: Ali # Output: 456 ■ 4.'?

A. Print

- B. instance
- C. access
- D. for

Explanation:

The correct answer, 'Print', is directly supported by the provided text: 'It is used to initialize attributes. Example: class Student: def __init__(self, name, roll_no): self.name = name self.roll_no = roll_no Creating Objects: s1 = Student("Ali", 123) s2 = Student("Sara", 456) Accessing Attributes: print(s1.name) print(s2.roll_no) # Output: Ali # Output: 456 ■ 4.'

Question No: 49 (Marks: 1)

Which term best completes the following statement: 'The self Keyword The self keyword refers to the current instance of the _____.'?

A. calc

B. Class

- C. attribute
- D. classes

Explanation:

The correct answer, 'Class', is directly supported by the provided text: 'The self Keyword The self keyword refers to the current instance of the class.'

Question No: 50 (Marks: 1)

Which term best completes the following statement: 'Dynamic Attributes Attributes can be added to an object at runtime. d.age = 5 _____(d.age) # Output: 5 However, this is not a common practice and may lead to inconsistent object structures. Notes for CS306 introduction to python Noor institute of technology ■ 8.'?

- A. inconsistent
- B. basic
- C. add

D. Print

Explanation:

The correct answer, 'Print', is directly supported by the provided text: 'Dynamic Attributes Attributes can be added to an object at runtime. d.age = 5 print(d.age) # Output: 5 However, this is not a common practice and may lead to inconsistent object structures. Notes for CS306 introduction to python Noor institute of technology ■ 8.'